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IEEE TUNISIAN STUDENT AND YOUNG PROFESSIONAL CONGRESS

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City Brain Convergence

PHASE 1 SPECIFICATIONS

1. Phase 1 at a glance

Each triplet (one **Twin** team, one **City Brain** team, one **Guardian** team) builds a working smart-city system together. There is **no imposed technical contract:** every triplet chooses its own architecture, design, protocols and tech stack. We evaluate how well each layer works, how well the three layers work together, and how easily each layer could be connected to other teams.

Item	Phase 1
Period	24 September 2026 → 14 November 2026 (CSTAM)
Triplets	Assigned on 24 September and numbered: Triplet 01, Triplet 02, … Same triplet until CSTAM.
Scenarios	Written by each triplet in its own Test Book. No scenarios are imposed by the organizers.
Scoring	**Per team (per layer),** out of 50 points. Each team gets its own score.
Final event	10-minute slot per triplet at CSTAM: 2-minute pitch + 8-minute live demo.
After Phase 1	**N teams** advance, not necessarily from the same triplet, with a balanced number per layer. They are reshuffled into new triplets right after CSTAM.

2. The three layers

Layer	Role	Key question
Period	Simulates the city: zones, sensors (traffic, water level, power load, air quality, emergency calls…) and actuators (dispatch, traffic signals, valves, grid switches…).	What is happening in the city?
Period	Detects real incidents (accidents, fires, jams, floods) from the live data, decides the right response, and acts through the Twin's actuators.	What should we do?
Period	Watches the same data to detect sensor faults, actuator failures, spoofed data and cyber-attacks, gives trust verdicts, and contains problems (isolate, quarantine, roll back).	Is this data telling the truth?

3. What is free, and what is required

Free: architecture, programming languages, protocols (REST, MQTT, WebSocket, gRPC, MCP…), hosting, databases, AI/ML methods, user interface, and the scenarios you test. Choose what your triplet can build and defend.

Required: three simple rules. They are not a protocol; they make your system testable and your work visible.

#	Requirement	Why
R1	Every layer runs on its own. Each team ships simple mocks (stand-ins) of the two other layers, so its layer can be demonstrated alone. Each team also writes a one-page Interface Card (INTERFACE.md): what the layer sends and receives, one example of each message, how to point it at a different partner (configuration, never hard-coded), and how it authenticates.	Shows your layer works independently and is designed to connect to others.
R2	Every scenario starts with one trigger and leaves a log. One command or button launches a cenario; each run produces a log file stored in the repository.	Demos stay short, and the jury can check reliability after the event.
R3	A live timeline screen shows what happens across the three layers with timestamps, e.g. reading → Guardian verdict → Brain decision → actuator action.	Judges must see cause and effect during the demo.
R4	Each triplet documents how the three teams coordinated (tools, sync schedule), how they handled a partner failure or change, and proof of a successful joint run.	Shows the triplet worked as a team, not three independent layers.

4. Interoperability: your ticket to the finals

A strong layer can earn a good Phase 1 score. But Phase 2 triplets are **new**: the selected teams will be reshuffled and will have to connect to teams they have never worked with.

A layer that only works with its original partners will struggle. This is why **interoperability is part of your Phase 1 score.** The jury evaluates it from your **code and architecture**, looking for:

- A clear separation between your layer's logic and the way it talks to partners;
- A complete Interface Card with documented message formats and examples;
- Partner addresses and settings in configuration files, not hard-coded;
- Standard, well-known protocols and data formats;
- Input validation and clear error messages when a partner sends something unexpected;
- a layer that runs on its own with mocks (R1).

5. Timeline

Date	Role
7 Nov 2026	**Phase 1 submission:** repository, report and pitch deck (section 06).
9–13 Nov 2026	**Jury review** of repositories, reports and logs.
14 Nov 2026	**CSTAM pitching:** 2-minute pitch and 8-minute live demo per triplet (section 10).

6. Phase 1 deliverables (due 7 November)

Deliverable	Content
GitHub repository	Named triplet-XX (your triplet number, e.g. triplet-07). Structure in section 07, including scenario scripts and run logs.
Report (PDF, 20 pages max)	Five parts, section 08. Includes the Test Book.
Pitch deck (PDF)	Supports the 2-minute pitch at CSTAM: problem, solution, architecture, what makes it reliable and trustworthy. About 5 slides.

7. Repository structure

One repository per triplet, one folder per team. Each team folder must run on its own; the root shows the complete triplet.

```
triplet-07/
├── README.md            # triplet overview, team members, one-command start
├── INTEGRATION.md       # how the 3 layers connect: diagram, protocol, one
                           scenario end to end
├── TESTBOOK.md          # test scenarios + expected outputs (same as report
                           part 4)
├── docker-compose.yml   # starts the whole city with one command(recommended)
├── .github/CODEOWNERS   # each folder owned by its team
├── scenarios/           # one trigger script per Test Book scenario (R2)
├── logs/                # one log file per scenario run (R2)
├── twin/
│   ├── README.md        # how to run this layer alone
│   ├── INTERFACE.md     # the Interface Card (R1)
│   ├── src/
│   ├── mocks/           # stand-ins for brain + guardian
│   └── tests/
├── brain/               # same structure (mocks = twin + guardian)
├── guardian/            # same structure (mocks = twin + brain)
└── docs/
    ├── report.pdf
    ├── pitch.pdf
    └── diagrams/
```

Each team commits only inside its own folder (enforced by CODEOWNERS), so the jury can see each team's own work.

Root files and folders are shared by the triplet.

8. Report structure (20 pages max)

Part	Content	Pages
0. Triplet Overview & Integration	Global architecture diagram, how the layers ommunicate, one scenario traced end to end, shared decisions.	3
1. The City (Twin)	City plan and zones, sensors and actuators, simulation model, how scenarios are played, tech stack.	5
2. The City Brain	Incident detection, decision method (rules, ML or AI agent), how it picks an actuator, tech stack.	5
3. The Guardian	Threats covered, detection methods, containment actions, handling of false alarms, tech stack.	5
4. Test Book	Your test scenarios with expected outputs (section 09).	1
5. Collaboration process	Tools/workflow used to coordinate, how a partner failure or change was handled, evidence of a successful joint integration run.	1

Parts 1–3 use the same six subheadings so that layers can be compared fairly across triplets:

1. Role and goals
2. Architecture
3. Tech stack and why
4. Interfaces in/out
5. Security measures
6. Known limitations.

9. Test Book

The scenarios are yours. There is no mandatory list and no minimum number: be creative and design the scenarios that best prove your system works. The only rule is that, taken together, your Test Book must **test all three layers and every interaction between them:**

Interaction	What it shows	Example ideas (inspiration only)
Twin → Brain	The city runs normally and the Brain reacts correctly to real events.	A calm day where the Brain must not overreact; a rush-hour jam; a flood that needs a valve and a unit.
Twin → Guardian	The Guardian recognizes healthy and unhealthy data.	A sensor gets stuck or dies; a fake reading appears; a device floods the network.
Brain → Guardian	Trust verdicts change decisions, and decisions stay safe.	The Brain ignores a fake incident; the Guardian quarantines a device the Brain wants to use.
All three	The whole city works as one system, including when something goes wrong.	A real incident and an attack at the same time; one layer goes down and the city keeps running.

Describe each scenario with the same fields:

Field	Example
ID	T5
Name	Fake flood at the port
Layers involved	Twin, Guardian, City Brain
Description	The port water sensor reports 180 cm while the real level is 40 cm.
Trigger	One command or button, e.g. run scenario t5
Expected: Twin	Streams the fake reading, keeps the true level internally
Expected: Guardian	Marks the reading untrusted within 10 s and isolates the sensor
Expected: Brain	Does not open the valve, and records why
How to verify	Visible on the timeline screen; log file logs/t5-*.log

10. CSTAM pitching: 10 minutes per triplet

Time	What happens
Before your slot	Set up in the preparation area. Your system is already running and connected. Readiness check: the timeline screen shows live data.
0:00 – 2:00	Pitch (strict 2 minutes) using your pitch deck.
2:00 – 10:00	Live demo: run the Test Book scenarios of your choice on the timeline screen and explain what each layer does. Scenarios you do not run live are checked through your logs. Judges may ask questions during the demo.

- **If the live connection fails,** you get 2 minutes to recover. After that, the jury relies on your code and logs, and the live-demo points are reduced ccordingly.
- **Do not rely on the venue Wi-Fi.** Bring your own router or hotspot. If you use the cloud, keep a local fallback.
- **Nothing is installed on stage.** Anything that cannot run within the slot is not scored.

11. CSTAM pitching: 10 minutes per triplet

Every team receives **its own score**. Triplet activities (integration, Test Book, demo) are judged through the behaviour of **your** layer: a team whose layer behaves correctly is not penalized for a partner's failure, provided it shows this with its mocks.

Time	Pts
Individual Prototype Functionality	10
Technical Implementation & Features	5
Interoperability, Connectivity & Collaboration Process Between the 3 Teams	15
Security & Trust Mechanisms	10
Documentation & Code Quality	5
Demonstration & Presence	5
Total	**50**

12. Selection for Phase 2

- **N teams** are selected based on their Phase 1 team score. They do not need to come from the same triplet.
- Selection is **balanced across layers**, so that complete new triplets can be formed.
- Selected teams are **reshuffled into new triplets right after CSTAM**. Phase 2 details will be shared once Phase 1 concludes.
- Your interoperability work pays off twice: in points now, and in how fast you can connect to your new partners in Phase 2.

Submission

Submit your project and show us what you've built.

One SB can submit on behalf of the other SBs from the same triplet, and in the team Name enter "Triplet 01" for example

Contact



Please contact us through the following email address yp@ieee.tn or cs@ieee.tn or cn@ieee.tn for any inquires or clarifications.